

FarmFederate

Federated LLM, ViT, and VLM for Privacy-Preserving Crop Stress Detection

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Short Technical Report

Paradigm	Modality	Best Model	Fed. F1	Fed/Cent
Federated LLM	Text (symptom text)	ALBERT-tiny	0.548	92.9%
Federated ViT	Image (crop photos)	ConvNeXT-tiny	0.659	86.1%
Federated VLM	Text + Image	CLIP fusion	0.785	92.6%

FarmFederate is the *first* framework to define, independently train, and directly compare three federated learning paradigms for multi-class crop stress detection, all trained under identical FedAvg conditions ($K=3$ clients, $T=8$ rounds, $E=3$ local epochs) with no raw farm data ever transmitted.

1. The Problem

Crop stress—encompassing drought, nutrient deficiency, pest infestation, disease, and heat damage—causes 20–40% yield loss globally. Existing AI solutions have two critical shortcomings:

- **Centralized:** farms must upload raw images or text to a shared server, violating data privacy and ownership rights.
- **Unimodal:** methods use images *or* text in isolation, missing complementary diagnostic signals.
- **No paradigm comparison:** no prior work trains Federated LLM, ViT, and VLM separately and evaluates which offers the best privacy-utility trade-off.

2. Three Independent Federated Paradigms

Each paradigm is trained and federated **independently**—the results are then compared directly to reveal their distinct operating points.

Paradigm 1 — Federated LLM

Input: Farmer-written symptom text (e.g., “leaves yellowing, dry cracked soil”).

Models: DistilBERT, BERT-tiny, RoBERTa-tiny, ALBERT-tiny, MobileBERT.

Model	Centralized F1	Federated F1
DistilBERT	0.571	0.530
BERT-tiny	0.585	0.543
RoBERTa-tiny	0.585	0.541
ALBERT-tiny (best)	0.590	0.548
MobileBERT	0.535	0.497

Key finding: Smallest federated accuracy gap (7.1%)—text features are *robust* to non-IID client partitioning. Ideal for bandwidth-limited or low-compute deployments.

Example: Input text: “brown spots on leaf margins, leaves curling upward, soil dry”
 Federated LLM prediction → **Disease Risk** (61% confidence)

Paradigm 2 — Federated ViT

Input: Crop photographs (224×224 RGB).

Models: ViT-Base, DeiT-tiny, Swin-tiny, ConvNeXT-tiny, EfficientNet (3-block residual CNN with batch norm + spatial dropout).

Model	Centralized F1	Federated F1
ViT-Base	0.696	0.599
DeiT-tiny	0.747	0.643
Swin-tiny	0.724	0.623
ConvNeXT-tiny (best)	0.765	0.659
EfficientNet	0.724	0.623

Key finding: Highest unimodal centralized accuracy (0.765) but largest federated gap (13.9%), indicating visual features are more sensitive to heterogeneous data distributions across farms. Future work: FedBN aggregation or pretrained ViT weights with LoRA fine-tuning.

Example: Input image: 224×224 crop photo — pale greenish-yellow leaves, drooping stems
 Federated ViT prediction → **Water Stress** (73% confidence)

Paradigm 3 — Federated VLM

Input: Text + Image combined via eight fusion strategies.

Fusion strategies evaluated:

Fusion	Centralized F1	Federated F1
Concatenation	0.797	0.738
Cross-Attention	0.807	0.747
Gated	0.779	0.722
CLIP (best)	0.848	0.785
Flamingo	0.816	0.755
BLIP-2	0.834	0.771
CoCa	0.783	0.725
Unified-IO	0.802	0.743

Key finding: Best absolute federated F1 (0.785) with a federation gap (7.4%) comparable to Federated LLM. CLIP contrastive alignment outperforms more complex Q-Former architectures. Multimodal gain: +0.083 over best unimodal (ViT), +0.237 over best text-only (LLM).

Example: Image: small holes in leaves, sticky residue Text: “*whitish deposits under leaves, plant stunted*”
 Federated VLM (CLIP) prediction → **Pest Risk** (84.8% confidence)

3. Dataset

The 5-class crop stress dataset is built from real HuggingFace corpora and synthetic imagery:

Class	Description	Visual Signature
Water Stress	Drought, wilting, stomatal closure	Vertical stripes, pale green
Nutrient Def.	Chlorosis, stunted growth	Concentric rings, yellowing
Pest Risk	Holes, sticky residue, webbing	Scattered dark spots
Disease Risk	Fungal/bacterial lesions	Coloured lesions, blight
Heat Stress	Leaf scorch, tip burn	Horizontal gradients, brown tips

Text: AG News + PubMed + SQuAD filtered for agriculture → ~1,089 balanced samples after rebalancing (original 25:1 imbalance corrected by capping majority at 2× median and oversampling minorities to median).

Images: PlantVillage + Beans (real) + synthetic class-distinctive patterns (pattern intensity 25–50%, 90% pattern presence, 20% cross-class confusion).

4. Anti-Collapse Mechanisms

Severe class imbalance (25:1) causes models to collapse to predicting only 1–2 classes. FarmFederate prevents this across *all three paradigms* via:

1. **Balanced Batch Sampling** — equal class representation per batch via minority oversampling using the maximum class count for batch size.
2. **Diversity Loss** — $\mathcal{L}_{div} = \lambda(1 - H(\bar{p})/H_{\max})$ with $\lambda=1.0$ and 70% entropy threshold; adds confidence penalty.

3. **Early Collapse Abort** — terminates training after 3 consecutive epochs where $>85\%$ of predictions fall into one class; restores best checkpoint.
4. **Focal Loss** — $\mathcal{L}_{focal} = -\alpha_c(1 - p_t)^\gamma \log p_t$ with $\gamma=2.0$ and class weights capped at $10\times$ (sqrt dampening: $\alpha_c = \min(\sqrt{n/Cn_c}, 10)$).

Result: 100% prediction diversity maintained across all 24 trained models and all three federated paradigms.

5. Key Results

Metric	Value	Notes
Best Federated F1 (VLM-CLIP)	0.785	$K=3, T=8$ rounds
Best Centralized F1 (VLM-CLIP)	0.848	Full data access
Avg. Fed/Cent Retention	90.5%	Across all 3 paradigms
Stress-biased Eval. F1	0.989–1.000	5 regional splits
Prediction Diversity	100%	All 24 models
Total Models Trained	24	$5 + 5 + 8 + 6$ variants
Hardware	NVIDIA Tesla T4	15.6 GB VRAM

6. Cross-Paradigm Comparison & Decision Guide

Paradigm	Cent. F1	Fed. F1	Gap	Best use case
Fed. LLM	0.590	0.548	7.1%	Bandwidth-constrained, text-rich farms
Fed. ViT	0.765	0.659	13.9%	Image-heavy, future FedBN aggregation
Fed. VLM	0.848	0.785	7.4%	Best overall; dual-modality available

7. Novelty

No prior work has:

1. Defined **Federated LLM**, **Federated ViT**, and **Federated VLM** as *three distinct federated paradigms*.
2. Trained each paradigm **independently** under identical FedAvg conditions.
3. Provided a **cross-paradigm comparison** quantifying the per-paradigm privacy-utility trade-off.
4. Applied any of these to **multi-class crop stress** (5 classes, not binary disease).

8. Practical Use Case

End-to-end scenario: A farmer in Punjab photographs yellowing maize leaves and types “*leaves pale yellow, lower leaves affected first, slow growth*” into the FarmFederate Android app.

The app encodes the image (ViT encoder) and text (LLM encoder) *locally*, fuses them via CLIP contrastive alignment, and sends only the **model update**—not the raw photo or text—to the federated server. After aggregating updates from three farms (Punjab, Kerala, UP), the global model predicts:

Nutrient Deficiency (78% confidence)

The app recommends: “*Apply 30 kg urea/acre. Re-evaluate in 10 days.*”

Privacy guarantee: The crop photograph never left the farmer’s device.

9. Conclusion

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- **Federated LLM** — most federation-robust (92.9% retention), lowest compute cost; best for text-rich, bandwidth-constrained deployments.
 - **Federated ViT** — highest unimodal centralized accuracy ($F1 = 0.765$) but most sensitive to non-IID heterogeneity; future work should apply FedBN or pretrained weights with LoRA.
 - **Federated VLM** — best overall federated performance ($F1 = 0.785$, 92.6% retention); CLIP fusion recommended when compute and dual-modality data are available.

FarmFederate demonstrates that the choice of federated paradigm is not merely a modality preference but a **fundamental design decision** with distinct privacy-utility implications, providing practitioners with clear, empirically grounded guidance for deploying federated agricultural AI.